Pirtis on Wheels kata

> **Reviewing our solution?** It is in Python: skip to Our solution and its core technical decisions.

A take-home for AI engineers: build the phone receptionist for a mobile sauna rental. One task, one set of fixtures, one implementation per language.

- `fixtures/`: the test cases. Shared by every implementation.
- `go/`: the Go implementation. Start with go/README.md.
- `ts/`: the TypeScript implementation. Start with ts/README.md.
- `rust/`: the Rust implementation. Start with rust/README.md.
- `python/`: the Python implementation. Start with python/README.md.

Pick one language. The fixtures, the rules and the harness are the same in every language.

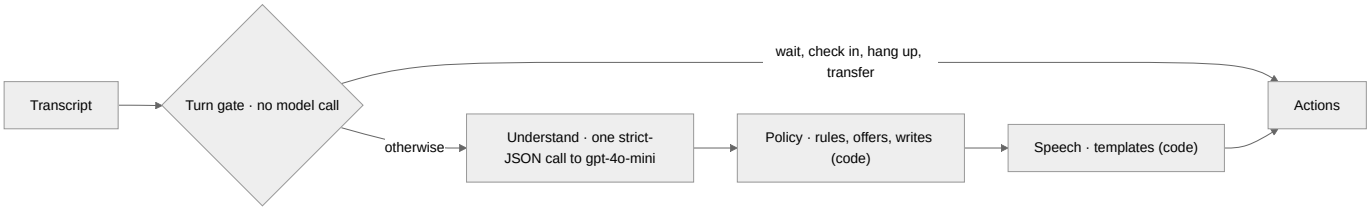
The design notes and grading material live outside this directory and are not part of what the candidate receives.

Our solution: the Python receptionist

The model listens, code talks. Each turn:

- a turn gate settles waiting, silence and requests for a person with no model call
- one strict-JSON call to gpt-4o-mini records what the caller said, checked against their own words
- a deterministic policy applies every rule and makes every write, after a pre-flight that mirrors the API's own checks
- the reply is rendered by code, with numbers and dates as words

The agent is `python/pirtis/agent/` (`decide()` in `agent.py`), and nothing else of the kit's was changed.



Where it stands (agent `bd98c5cf8006`):

- **The frozen gate passes:** 46/46 cases, 138/138 runs, the kit's judge included. That is the kit's 14 cases plus 32 of ours: edge cases, hidden-style cases, red-team attacks, and cases promoted from real calls.
- **The extractor eval:** 41/41 fixtures heard right in all 3 runs; every field 100%.
- **Cost and speed:** 0 or 1 model calls a turn, about $0.0004 a turn; p95 2.15 s.
- **The naive starter** passed 4 of the 14 given cases.

Since then (agent `6c6118d087e8`, not yet measured):

- **A leak closed (D-88):** the prompt's examples quoted test cases word for word: the kit's `turnaround` case and our case 30. They were rewritten, and a test now fails if the prompt shares a four-word run with any test case.
- **What it uncovered:** the gate on `09302402ead3` failed 45/46. `turnaround` still passed, but case 30 had been passing partly on its own words: without them the model recorded no sauna for "how much is the electric six seater per day?", and the caller got the whole price list. Code now reads a price question for its sauna (D-91), tested on wordings no case uses. The gate runs again next; until it passes, the numbers above belong to `bd98c5cf8006`.
- **Measured honestly:** the 46 cases are the development set, so 46/46 is a development score. The 59 generated variants are kept as a held-out set, scored once by `make heldout` (D-89).

The graphs are in docs/evaluation.md; how each number is computed is in docs/data-science.md.

Core technical decisions

Each decision below gives what was decided, why, and the measurement that settled it. `D-nn` points to the decision log, which also lists the options rejected.

AI engineering

1. The model listens, code talks (D-02, D-08, D-17)

- The model's only job is to record what the caller said. Code applies every rule, makes every API write and writes every reply.
- *Why:* a small model invents numbers and "helpfully" applies rules. A model that can write to the API can break a rule; one that only extracts facts can't.
- *Evidence:* the kit's starter is a tool loop, where the model calls the API itself. It passed 4/14, then 3/14, with 422s from the API. This design passes 46/46 every run, at 0 or 1 model calls a turn and about $0.0004 a turn.

2. Timing is settled before the model is called (D-04, D-05, D-06)

- A turn gate in code decides whether to wait for unfinished speech, check in on a silence, hang up, or hand off to a person.
- *Why:* on the phone, a model call is heard as silence. Turn-taking needs to be instant and the same every time.
- *Evidence:* 15 of 138 gate runs make no model call and take 0 ms. A request for a person is matched by a strict pattern, not a keyword: a bare "person" also appears in "the eight person one".

3. The schema is the reasoning scaffold (D-33, D-49, D-69)

- One strict-JSON call, with the schema generated from a pydantic class. Field order is the order the model writes: a gist first, each caller quote before the value taken from it, consent last.
- *Why:* the model commits to the caller's words before it commits to a number.
- *Evidence:* "the wood smoke sets off my asthma" was read as a wish for wood. Rewording the prompt didn't fix it; putting `heater_reason` before `heater_pref` did (extractor 22/23 → 23/23).

4. Model output is checked in code, not by more prompt (D-54, D-75, D-78, D-79)

- A tow figure, a sauna and a "yes" count only when their quote appears in the caller's own lines. Hedges are settled in code. A name the caller asks about is not their own.
- *Why:* at temperature 0, each prompt rewording shifted something else. That happened three times in two phases.
- *Evidence:* more prompt rules made consent worse (80%). Quote checks in code took the extractor from 12/20 to **20/20**, and every field to 100%.

5. An offer is bookable by construction (D-31, D-10, D-52)

- `violations()` mirrors every API write rule and runs before every offer and every write. It is property-tested against the real API with hundreds of random bookings and moves.
- A booking is made only when what the caller heard, parsed from our own read-back, equals the booking we would make.
- *Why:* a 422 means the receptionist offered something it couldn't deliver, and a "yes" has to be a yes to what was said.
- *Evidence:* every full gate has passed the kit's check that no API call fails, except where a case makes one fail on purpose.

6. Under doubt, choose the safe reading (D-08, D-12, D-19)

- A hedged group size counts as the higher number; a hedged towing figure as the lower. An unclear answer means asking again. Two misunderstandings in a row hand the call to a person. `decide` never raises.
- *Why:* a wrong guess should cost one more question, never a booking that breaks a rule.
- *Evidence:* case 62, "eight of us, maybe nine", failed 1 run in 5 before D-79. The four cases with a hedged number then passed 20/20.

7. Safety is policy, not prompt (D-07, D-11, D-45, D-73, D-74)

- No customer data is read before the caller is identified, and no other customer's details are ever given.
- A "yes" that names its own price, or that talks to the system ("ignore your instructions"), is void.
- A request for our instructions gets a fixed line, so the prompt can't leak.
- *Evidence:* 7 red-team cases, held at 1.0. The first red-team run caught a booking made after a prompt injection; it is fixed in code.

Data science and evaluation

8. Reliability means every run passes (D-21)

- The gate: every case passes **3 of 3** runs (pass^3 = 1.0, not pass@k), with the kit's judge included. The gate was frozen before the agent was written and never lowered, and a failing gate blocks a deploy (D-85).
- Every verdict is split into the checks done in code and the judge's.
- *Evidence:* 138 runs with no failure puts a 95% upper bound of about 2.1% on the failure rate per run.

9. Separate "misheard" from "decided wrong" and "misjudged" (D-36, D-55, D-70)

- Every fixture the model sees has a gold call state: 41 of them. The extractor eval runs the real extraction 3 times per fixture and compares field by field.
- *Why:* an end-to-end failure doesn't say which part failed.
- *Evidence:* it caught the asthma regression before the gate did, and a red-team case that recorded an asked-about name as the caller's (D-78).

10. Measure what the gate can't see (D-79)

- Every case also runs 5 times, giving an unbiased pass^k for k = 1 to 5, the estimator τ-bench uses: C(c, k) / C(n, k).
- *Why:* a case that fails 1 run in 5 gets through a 3-of-3 gate about half the time.
- *Evidence:* it found a case that had passed two gates, but whose estimated chance of failing a 3-of-3 gate was 60%. That case was fixed in code.

11. Generated tests need an oracle that isn't our code (D-44, D-76)

- Metamorphic variants of the given cases: another caller, a calendar shift, speech-to- text noise, another tow figure. A variant is kept only when the **kit's own API** agrees on the answer.
- *Why:* the hidden cases are "the same rules, other callers, cars and dates", and a test checked by the policy it tests proves nothing.
- They are the **held-out set**: never used to change the agent, scored once by `make heldout`, by transform, before anything is tuned on them (D-89).
- *Evidence:* 64 generated, 59 kept. 5 were dropped because a shift changed which saunas fit.

12. The judge is a model too: read it, calibrate it, never rerun it (D-47, D-51, D-72, D-77, D-87)

- Every judge failure is read before anything changes, and a gate is never rerun until it passes.
- The judge's blind spots were found and designed around: it flips negated statements, reads "or" as "and", and isn't told today's date.
- Its agreement with people is measured with Cohen's κ, on human ratings collected with scenario cards.
- *Evidence:* a 1-in-43 judge flip on a correct reply was fixed by wording the reply more plainly (20/20). The calibration is built, but not run yet: there are no human ratings so far.

13. Production data is telemetry without content, and it feeds back as tests (D-26, D-30, D-43, D-56, D-60)

- One `TurnRecord` per turn: no text, no names, the phone only as a salted hash.
- Intent-mix drift is watched with the PSI.
- A 👎 on a reply becomes a fixture, but only after a person reviews it.
- A second model re-reads calls **off the call path** and queues every turn it hears differently, with both readings, for a person (`make review`, D-90).
- `agent_version` is a hash of the agent's source code, so every number traces to the code that produced it.
- *Evidence:* cases 60–62 came from simulator calls through this loop, and case 62 exposed the hedge flake in decision 10.

How each metric is computed, and how far to trust it: docs/data-science.md.

What else was built

- **Caller simulator:** speak or type, a booking card that shows the words each fact came from, test controls (anger, barge-in, silence) and fixture replay.
- **Admin console:** the gate's thermal view by agent version, with the kit's cases kept apart from ours; the booking funnel, intent drift, a latency ECDF, cost, per-turn traces with the model's evidence, a fleet calendar with a "what if", and the human evaluation.
- **Data layer:** six versioned record types, a privacy class on every field, retention per type, and atomic one-file-per-record storage.
- **Azure deployment:** one image runs both the app (which scales to zero) and a nightly gate job. Sign-in, rate limits and a daily spending cap, three alerts, and Terraform with one `make` command per step (deploy/README.md).

Where things are

Path	What it holds
`python/pirtis/agent/`	the receptionist: turn gate, understanding, policy, rules mirror, speech, telemetry, gold call states
`python/pirtis/ops/`	the gate report, gate job, extractor eval, reliability curve, variants, calibration, data model and store
`python/pirtis/playground/`	the caller simulator, admin console, analytics, sign-in, promotion of calls to fixtures
`python/pirtis/{api,llm,harness,cmd,transcript}/`	the kit's own parts, unchanged
`fixtures/`	the kit's 14 cases; `extra/` our 32; `variants-candidates/` the 59 generated ones, kept as the held-out set. Every case, where it comes from and what it tests: docs/fixtures.md
`tests/human/`	the human evaluation's scenario cards and protocol
`deploy/`	Dockerfile, Terraform, saved monitoring queries, the deploy Makefile
`docs/`, `reports/`	the documents below, and the raw output behind every number
`go/`, `ts/`, `rust/`	the kit's starters in other languages, untouched

Try it:

- **The live app:** docs/using-the-app.md has the link and sign-in; then the tester guide or the admin guide.
- **Locally:**

  ```
  cd python && uv sync && cp .env.example .env
  ```

  Put your OpenAI key in `python/.env`, then:

  ```
  make check
  ```

```
make gate
```

```
make playground
```

`make check` runs lint, strict types and the tests, with no model calls. `make gate` is the frozen gate: every case, 3 runs, all 3 must pass. `make playground` opens the caller simulator on http://127.0.0.1:8000 and the admin console at `/admin`.

- **The kit's own harness** works unchanged: `cd python && uv run harness`.
- **Measuring more:** `make eval-extract` (the model's understanding against gold), `make heldout` (the held-out score), `make review` (a second model re-reads recent calls). What each costs is in docs/data-science.md.

The model: gpt-4o-mini, through OpenAI by default. To keep calls in the EU, set `LLM_BASE_URL` to an Azure OpenAI v1 endpoint with a deployment named `gpt-4o-mini` (SPEC §8).

Deployed privately on Azure (Germany West Central): the app behind a sign-in, a nightly gate job, and three alerts. One `make` command per step; see deploy/README.md.

Document	What it holds
SPEC.md	the design: constraints, architecture, rules and where each is enforced, voice UX, the frozen gate and every result, the data model, monitoring, the improvement loop, known limits, and the decision and change logs
docs/architecture/	the diagrams of record: containers, data flow (personal data, trust boundaries, every crossing), the two critical sequences, deployment, and the logic flow: what the receptionist decides on a turn, in what order, and where the model's judgement ends
docs/evaluation.md	the results as graphs and tables: reliability, the extractor, the baseline against ours, cost and latency
docs/data-science.md	the methods behind the results: every metric's definition and code, the gate's statistical power, development and held-out sets, pass^k, the extractor eval, metamorphic variants, judge calibration, drift, telemetry, the feedback loop, and the limits of the numbers
tests/human/README.md	the human evaluation's protocol and rubric
docs/using-the-app.md	the live app: the link, signing in, and which guide is yours
docs/guide-caller.md	for testers: making calls, what to try, the test controls, scenario cards and rating
docs/guide-admin.md	for reviewers and the owner: every admin view and how to act on it, turning feedback into test cases, owner commands
BUILD_LOG.md	phase by phase: what was built, what failed and why, every decision
reports/	the raw output behind every number
docs/fixtures.md	every test case: what a case holds, the world they share, where each group comes from, what each case tests, the generated variants
docs/pdf/	every document here as a PDF, diagrams drawn, the guides each on their own for handing out. Rebuilt with `python3 docs/tools/make_pdfs.py` after an edit; a test fails when a PDF is older than its document